Time for transparent targeting: an investigation of targeting disclosures, coping mechanisms, credibility, and political attitude

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Abstract

A major problem with political microtargeting (PMT) is that users are often unaware that they are being targeted, whereas current transparency approaches fail to inform users. Therefore, the current work investigates two types of disclosures on a political advertisement, one that is currently used by platforms: a light gray sentence stating that content is sponsored, and a salient and dynamic disclosure that informs users about the parameters and data points used to target them. This work investigates whether these disclosures, as a means of transparency to counter the potential negative effects of PMT, affect users in terms of persuasion knowledge, resistance, source credibility, and attitude towards the advertised politician. In a preregistered online one-factorial between-subjects experiment ($N = 547$), we found no evidence for the claim that a more elaborate, salient targeting disclosure increases users' persuasion knowledge. However, our results show that persuasion knowledge is related to cognitive resistance, which, in turn, is related to both source credibility and users' attitude towards the advertised politician. Our results further indicate that this process is cognitive, rather than affective. Finally, most participants seemed to recall a disclosure, although it was not always the correct one.

Keywords: microtargeting, disclosures, resistance, credibility, persuasion knowledge

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As people become more active online, the landscape of political communication shifts with their target audiences. Large parts of communication between (potential) voters and politicians occur online, specifically on social networking sites (SNS) (Giasson et al., 2019). Online behavior leaves digital breadcrumbs, especially on SNS, where behavior is interactive, as opposed to websites where most people consume information and a more one-directional sender-receiver relationship exists. The digital traces that users leave behind are very valuable to political consultancy firms, which use these data points to segment small groups of users and narrowly target them with political advertisements (Gandy, 2000; Murray and Scime, 2010). The allocation of users into groups and adapting messages that are supposed to resonate effectively with a specific audience is called *microtargeting* or, in the political landscape, *political microtargeting* (PMT) (Kosinski et al., 2013; Kruikemeier et al., 2016; Matz et al., 2017; Zarouali et al., 2020; Zuiderveen Borgesius, 2016). Two prime examples of the use of PMT are the 2016 United States elections and the Brexit referendum in the United Kingdom. In both examples, political consultancy firm *Cambridge Analytica* presumably used PMT through the gathering and connection of data from more than 50 million Facebook users, building psychological profiles, and targeting those users with messages made to resonate with them (Cadwalladr, 2018). In addition, PMT can be used not only to persuade voters, but also to discourage political participation, including electoral turnout (Bodó et al., 2017). One of the problems with PMT is that people do not recognize persuasive attempts that senders use to affect them. One solution to this could be the use of disclosures. These labels inform users directly about the content they read or engage with, showing that content is sponsored or paid for by certain parties or companies (e.g., a collaboration between an influencer and a brand). In the case of PMT, disclosures provide platforms with the

possibility of informing users that they are targeted by the sender of a message through their data.

Although research on marketing and advertising has already focused on disclosures (Amazeen and Wojdynski, 2020; Boerman et al., 2015; Cain, 2011), research on disclosures for political microtargeted advertisements is still scarce (Binford et al., 2021; Kruikemeier et al., 2016). In addition, the effects of seeing disclosures could go beyond simply informing receivers and prompt psychological mechanisms, such as counterarguing and contesting the message, but can also negatively affect users' image of the party sending the message or the politician being advertised (Kruikemeier et al., 2016; van Reijmersdal et al., 2016). However, currently used disclosures do not always seem to be correctly recalled (Binder et al., 2022; Binford et al., 2021; Kruikemeier et al., 2016).

Therefore, the current work investigates two different types of disclosures: a regular sponsorship disclosure and a dynamic targeting disclosure containing more information. In this disclosure, users can hover over a targeting measure to show the parameters used to target them (i.e., age, gender, and behavior). Building on and expanding existing work, this study has an experimental design and focuses on the two different types of disclosures. Consequently, we aim to examine how users cope with messages when it is disclosed that they are targeted by the sender of the message, and the mechanisms that are activated after exposure to the disclosure, being persuasion knowledge, and cognitive and affective resistance. Moreover, we investigate the different outcomes of these mechanisms with regard to source credibility and attitude towards an advertised politician.

Theoretical background

One of the problems regarding microtargeting is that it is a black box, and neither laypersons, journalists, nor scientists know exactly what is done by political consultancy firms. This black box makes it difficult to research and practically impossible to estimate the

effects of PMT (e.g., Cambridge Analytica's practices). Since the lack of transparency and information about the algorithms and models of political consultancy firms combined with the lack of a regulatory body makes it hard to focus on the 'sender' side of PMT, the solution could be to inform users by implementing transparency measures.

Disclosures

One approach concerning transparency and informing users could be the use of disclosures, specifically targeting disclosures (Binder et al., 2022; Binford et al., 2021; Jansen and Krämer, 2023; Kruikemeier et al., 2016). Disclosures on posts and advertisements on SNS were initially used to inform users of the messages being sponsored, labeling them as such. Within research on marketing and advertising, work already exists on disclosures, showing that they facilitate the recognition of advertisements (Amazeen and Wojdynski, 2020; Boerman et al., 2012, 2015; Jung and Heo, 2019).

Research on disclosures regarding PMT seems to be emerging but is still scarce and shows mixed results concerning recognition (Binder et al., 2022; Binford et al., 2021; Jansen and Krämer, 2023; Kruikemeier et al., 2016). Recent work acknowledges the opportunities that disclosures bring to inform receivers of the targeting practices that occur. Following the EU Digital Services Act (DSA), targeting disclosures should overlap with sponsored disclosures in containing information on when and on whose behalf content is displayed but simultaneously go beyond that and include individualized data used to target users and parameters used for those data points (European Commission, 2022).

Regarding the design of disclosures, the ones that SNS use nowadays, light grey embedded around the content (e.g., 'Sponsored, paid for by …'), seem to be insufficient to inform users and contribute to transparency (Binder et al., 2022; Binford et al., 2021; Kruikemeier et al., 2016). However, more prominent disclosures seem to be effective in informing users of the persuasive or targeted nature of a message. Prominent disclosures are

those that are more easily seen by users and therefore lead to higher levels of recognition that an advertisement is an advertisement (Amazeen and Wojdynski, 2020) For instance, research shows higher levels of advertising recognition with a highlighting square around the ad and the disclosure (Jost et al., 2022). Other work suggests that disclosures that include information that aligns with the regulations in the DSA are noticed more easily (compared to disclosures investigated in prior advertising research) (Dobber et al., 2023). In line with this, the current work investigates a more extensive disclosure containing information about the advertisement being targeted at users. As an exploratory feature of the disclosure, we implement a dynamic function showing participants that the advertisement is targeted to them based on their information. This makes the disclosure used in the experimental condition align more with the DSA, which states that the parameters and data points used to fill in those parameters should be disclosed (Digital Services Act, 2022; European Commission, 2022). This type of disclosure has been shows to lead to better levels of disclosure recall (Dobber et al., 2023). Besides providing participants with information that fits their person, the dynamic feature might also lead to better cognitive absorption and a more positive attitude towards the content (Oh and Sundar, 2015). In addition, the usage of red and underlined text for the parameters within the disclosure not only shows participants that extra information lies behind the word (e.g., as is the case with a hyperlink on a website) but the color red also attracts more visual attention (Baik et al., 2013), while making the disclosure more prominent since it stands out more in a user's timeline on a platform (Amazeen and Wojdynski, 2020). Moreover, the hover feature can show participants information in a more fluent manner than Facebook's 'Why am I seeing this ad' because the popup will (dis)appear based on the mouse movements and is therefore less interrupting and more readily available, while still being applicable in the platforms corporate design, compared to large boxes drawn around an advertisement (Jost et al., 2022). While existing work underlines the importance of

investigating new types of disclosures that adhere to the regulations of the DSA (Dobber et al., 2023; Jost et al., 2022), we examine a newly developed targeting disclosure with red and underlined text, as well as the option to see the actual datapoints that were used regarding the targeting parameters. To the best of our knowledge, targeting disclosures such as these have not been studied with regard to PMT.

Persuasion knowledge

Nowadays, users of social networking sites (SNSs) are being exposed to an increasing number of persuasive messages. To cope with these messages and their potential influence, receivers develop beliefs and knowledge regarding these attempts through previous exposure and experience. The overarching term most commonly used to describe these mechanisms is *persuasion knowledge* (Friestad and Wright, 1994). Research shows that if messages contain a disclosure, persuasion knowledge is more likely to be activated, whereas if there are no disclosures on a message, persuasion knowledge is less likely to be activated (Boerman et al., 2012; Kruikemeier et al., 2016; van Reijmersdal et al., 2016). However, recent work emphasizes the importance of users actually seeing and perceiving the disclosure, while only then can the disclosure inform them and activate persuasion knowledge (van Reijmersdal et al., 2023). Moreover, seeing a disclosure can also lead to a change of meaning of the message (i.e., being aware of persuasive intent), which in political advertising could mean that people think the message might be less sincere (Friestad and Wright, 1994; Wojdynski and Evans, 2020). A previous study found a positive relationship between a disclosure and persuasion knowledge in the PMT context (Kruikemeier et al., 2016). However, this study used a sponsored disclosure and training condition, but not a dynamic disclosure. Moreover, other work shows that including parameters in the disclosure, as is obligated by the DSA, could be promising regarding recall of the disclosure as well as persuasion knowledge (Dobber et al., 2023).

As the practices of PMT differ from those of regular political advertising, we consider it important to examine whether the use of PMT, when recognized, generates differences in effects compared to regular political advertising. Furthermore, we propose that activating persuasion knowledge is the first step for users to cope with persuasive messages. Persuasion knowledge has been found to be a valid measure to scrutinize users' knowledge of the persuasive tactics occurring. However, the extension of this measurement applied to a microtargeting context developed by Binder and colleagues (2022), *targeting knowledge*, has not yet been investigated in other research. Therefore, as an exploratory research question, we compare our measured constructs between our two experimental groups, and also compare the level of targeting knowledge between those groups. Thus, we can compare persuasion and targeting knowledge between our groups, and scrutinize whether these constructs differ between and within groups.

Current research acknowledges the activation of persuasion knowledge through the implementation of disclosures (Binder et al., 2022; Dobber et al., 2023). However, disclosures such as those in the current study have not been investigated yet since the disclosures are newly developed. We expect that our targeting disclosure, including interactive parts that show the actual parameters used to target the user, activates higher levels of persuasion knowledge than a regular disclosure stating that the message is sponsored:

H_1: Exposure to a targeting disclosure leads to higher levels of persuasion knowledge than exposure to a sponsored disclosure.

RQ_1: What are the differences between users who receive a sponsored disclosure and users who receive a targeting disclosure regarding persuasion knowledge and targeting knowledge?

Resistance

Youn and Kim (2019) found that when an advertisement is recognized as such, persuasion knowledge is activated, and in turn, resistance could be activated as a coping mechanism, which negatively affects users' attitudes towards the content. Furthermore, research shows that when persuasion knowledge is activated as a coping mechanism for persuasive attempts, people use their existing knowledge to either be persuaded or resist the message (Sagarin et al., 2002; Tormala and Petty, 2004; van Reijmersdal et al., 2016). According to Quinn and Wood (2004), resistance is the process through which one rejects influence and retains one's views. We propose that the activation of persuasion knowledge leads to resistance because receivers feel that a message is a perceived threat to freedom (Dillard and Shen, 2005). This potential threat of freedom can be countered through the activation of resistance, increasing users' certainty of their initial attitudes (Tormala and Petty, 2002). Furthermore, studies on online behavioral advertising, which can be seen as the predecessor of PMT, have found that the technique has been perceived as intrusive (Segijn and van Ooijen, 2022) and even creepy (Segijn et al., 2022). This could mean that if users are made aware of the targeting practices taking place through disclosures and recognize this through persuasion knowledge, they tend to resist the message more. However, to the best of our knowledge, the concept of resistance to persuasion has not been investigated in the context of microtargeting.

People's attitudes contain distinctive cognitive and affective components. These components have valence dimensions that range from negative to positive (Ostrom, 1969; Rosenberg et al., 1960). Persuasion, which can be considered an attempt to change existing attitudes towards a brand or, in the case of PMT, a politician, has also been found to have different cognitive and affective dimensions (See, 2018), which makes receivers react differently to stimuli depending on which dimension is activated more (Di Plinio et al.,

2022). In addition, attitude change can be a function of one's favorable cognitive and affective reactions to a message simultaneously (Petty and Cacioppo, 1986; Rosselli et al., 1995). Furthermore, cognitive responses can mediate the effect of exposure to a message on a persuasive outcome more than affective responses can (Banerjee and Greene, 2012).

Users' resistance to a message can also be both cognitive ("I do not believe it!") and affective ("I do not like it!"), also at the same time (Knowles and Linn, 2004, p. 7; Sagarin and Cialdini, 2004). Previous studies on sponsored content and advertising show that affective and cognitive resistance can be activated as a result of persuasion knowledge activation (Fransen, Smit, et al., 2015; Fransen, Verlegh, et al., 2015). We deem it important to investigate both routes simultaneously and scrutinize potential differences in resistance and differences in psychological effects on users. Cognitive resistance towards a persuasive message means that users have a negative reaction towards a persuasive message based on their preexisting thoughts and various (message) factors (e.g., disclosure or through recognizing it themselves). Previous work has found that counterarguing is an effective and often-used strategy for resisting messages or advertisements (van Reijmersdal et al., 2016; Zuwerink Jacks and Cameron, 2003). Additionally, disclosures could increase thinking about advertising and decrease positive or neutral cognitions about brands or advertisers, thereby activating cognitive resistance (Boerman et al., 2012). To the best of our knowledge, resistance in these forms has not yet been investigated in the case of microtargeting. Building on the work of van Reijmersdal and colleagues (2016), which distinguished between cognitive and affective resistance with regard to disclosing sponsored content in blogs, we adapt this to the context of political microtargeting and expect that persuasion knowledge has a positive relationship with the cognitive dimension of resistance:

H_2: Persuasion knowledge is positively related to cognitive resistance.

Although different kinds of cognitions as a response to persuasive messages are important, users also tend to have affective responses to these messages. An affective response can be defined as a change in a user's mood or feelings after exposure to a message (e.g., getting annoyed or irritated by commercials) (Edell and Burke, 1987; Schwarz et al., 1991). A positive mood reduces the motivation to process content and contextual signals systematically. Equivalently, a negative mood induces more effortful, detail-oriented, analytical, and critical processing, which could lead to skepticism towards the persuasive message (Boerman et al., 2012; Bohner et al., 1992). Other studies have shown that persuasion knowledge activation is positively related to affective resistance with regard to disclosing sponsored content in blogs (van Reijmersdal et al., 2016) and can even prompt feelings of anger (Beckert et al., 2021). Examining the second route of resistance following van Reijmersdal and colleagues (2016), we expect that in the case of microtargeting, persuasion knowledge has a positive relationship with the affective dimension of resistance:

H_3: Persuasion knowledge is positively related to affective resistance.

Source Credibility

Research on advertisement disclosures on SNS shows that disclosures are likely to have a negative effect on the credibility of the source of a message and the message itself (Carr and Hayes, 2014; Deng et al., 2020; Stubb and Colliander, 2019; Wojdynski and Evans, 2016). Source credibility is mostly conceptualized as a set of positive characteristics associated with the source of a message that could potentially increase the acceptance of the message by receivers (Housholder and LaMarre, 2014; Ohanian, 1990). Receivers' perceived credibility influences how they use and process the information that they receive from a source (Madsen, 2019).

Moreover, credibility has been found to lead to more positive responses to persuasive attempts, as described in the Elaboration Likelihood Model (ELM) (Petty and Cacioppo, 1984). In addition, credibility is an important factor when the subject of discussion is something that the receiver of a message has less knowledge about, which is often the case for political topics (Yoon et al., 2005). Source credibility is an important predictor of voting intention, and a decreased perceived credibility tends to decrease the willingness to vote for a certain party or candidate, possibly making it counterproductive for political actors or parties to take actions that potentially lower credibility (Hetherington, 1999; Housholder and LaMarre, 2014; Madsen, 2019). Although disclosures are usually obligatory by regulations or laws, and not something political actors and parties choose to implement, they could still benefit from the awareness of potential negative effects (European Commission, 2022).

In line with van Reijmersdal and colleagues (2016), we expect that if people are informed about a message being sponsored or targeted at them through a disclosure, they activate resistance through persuasion knowledge. Subsequently, we propose that higher levels of resistance decrease the perceived credibility of the source (Carr and Hayes, 2014). In this study, we use the construct of source credibility as a measure to research the credibility perceptions that users have towards the political party that sends the message. While the politician who is advertised might sometimes be the source of the message, we used the party, which is also shown to be the sender of the ad in the stimulus material. We propose that being aware of the targeting that a source employs to persuade users has the potential to increase resistance and lower users' perception of credibility regarding the source:

H_4: Cognitive resistance is negatively related to source credibility.

In line with our other hypotheses and earlier work, we investigate the two separate routes of cognitive and affective resistance, as they are two separate psychological processes. The rationale behind measuring the routes as separate is a potential difference between counterarguing with a message or sender or actually getting annoyed or irritated. There are mixed results on the activation of cognitive and affective resistance at the same time (Fransen, Smit, et al., 2015; Petty & Cacioppo, 1986; Rosselli et al., 1995) and separately (Banerjee and Greene, 2012; Di Plinio et al., 2022; See, 2018), and their effects on persuasive outcomes. Therefore, we scrutinize both types of resistance and propose that, like cognitive resistance, affective resistance is negatively related to source credibility and investigate the potential differences between the types of resistance:

H_5: Affective resistance is negatively related to source credibility.

RQ_2: What are the differences in the relationships between cognitive and affective resistance, and source credibility?

Attitude towards the politician

When receivers are exposed to content on SNS, they form attitudes towards both the content and the sender. However, the sender of the content (source) does not always have to be the person displayed (e.g., a post from a political party, advertising a specific politician). Users' perceived credibility of a source has been found to be one of the factors influencing their attitudes towards that source (Kumkale et al., 2010). Furthermore, since source credibility has been shown to predict voting behavior and choice for a candidate (Madsen, 2019, p. 107), and high levels of source credibility have been shown to predict more positive candidate evaluations, we deem it important to research a possible relationship between the perceived credibility of the source and receivers' attitudes towards this source (Funk, 1999; Markus, 1982; Miller et al., 1986). Moreover, in earlier work on disclosures and brand

attitudes, van Reijmersdal and colleagues (2016) found that cognitive and affective resistance have negative relations to attitudes, in line with work that found that counterarguing a persuasive message leads to lower persuasion in terms of attitudes (Zuwerink Jacks and Cameron, 2003). Furthermore, reactance to a message has been found to have a negative effect on brand attitudes (Beckert and Koch, 2022). In line with this, we propose that users could resist a message containing a disclosure because it feels intrusive to be targeted, which lowers users' attitude towards the politician:

H_6: Cognitive resistance is negatively related to attitude towards the politician.

H_7: Affective resistance is negatively related to attitude towards the politician.

RQ_3: What are the differences in the relationships between cognitive and affective resistance and attitude towards the politician?

Furthermore, higher levels of source credibility lead to more favorable attitudes than lower levels of source credibility (Tormala et al., 2006). Accordingly, we propose a positive effect of source credibility on users' attitude towards a politician:

H_8: Source credibility is positively related to attitude towards the politician.

Finally, as an exploratory research question, we investigate the potential differences between our two different types of disclosures on cognitive and affective resistance, source credibility, and attitude towards the politician. This enables us to at least speculate about the most fitting designs concerning disclosures on microtargeted advertisements on SNS, which is important not only to inform users but also for platforms to oblige to the new guidelines from the European Union.

RQ_4: What, based on exposure to the different types of disclosures, are the differences in cognitive and affective resistance, source credibility, and attitude towards the politician?

An overview of our proposed hypotheses and path model is shown in Figure 1.

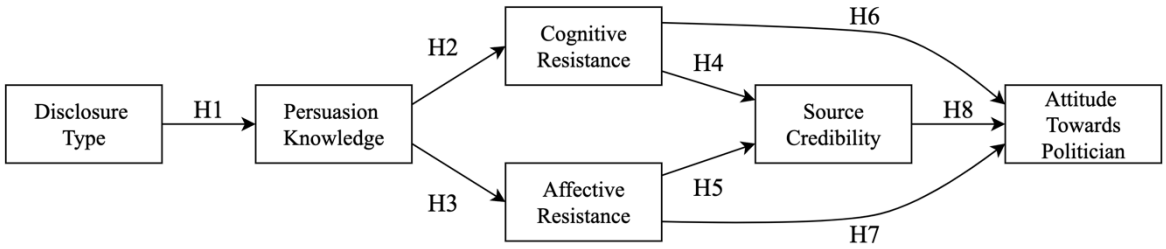


Figure 1

Proposed path diagram

Method

The online experiment was approved by the ethics committee of [Anonymized]. We preregistered this study before collecting data: https://osf.io/sftbe/?view_only=10057e05315342cbbae9e95e91c2da20. All participants provided informed consent before participation. Supplementary materials, code, and measures in both English and German are publicly accessible on OSF: https://osf.io/dwygm/?view_only=9e486925981f4b72ab9c16e59369687c

Design

To test our hypotheses, we conducted an online experiment using a factorial between-subjects design containing two groups. Participants were exposed to a Facebook advertisement containing either a sponsored disclosure, in line with the disclosures Facebook uses ("Sponsored, paid for by …") in the control condition, or a Facebook advertisement with a targeting disclosure ("This message is targeted on you based on your: age, gender, and online behavior"). The reason that there is a sponsored disclosure on the advertisement in our

control condition is that an advertisement on Facebook without a disclosure would be an organic post and not an advertisement because these disclosures are regulated and mandatory.

When hovering over age, gender, and online behavior on the targeting disclosure, more information about the targeting parameters was provided. These parameters were based on the demographic information we asked the participants before the start of our microtargeting simulation and were thus in line with their answers. Participants' registered age was converted into groups (e.g., a participant that filled in their age as 35, fell into the 31-40 years group) and the following message was given: *This message is targeted on people between the ages of [age group]*. The gender parameter was a direct insert of the gender the participant answered that they identified with (female, male, or diverse), and the following message was given: *This message is targeted on [gender] users.* For online behavior, we were unable to use information, as we did for demographics. Therefore, the following message was given: *This message is targeted on you through your online behavior and interactions on Facebook.* We were able to measure how many times participants hovered over each term individually, and thus, were exposed to the targeting parameters we included. We recognize that the hover option was possible because participants took part in the experiment on a computer. On a smartphone or tablet, participants would need to click on the term, which we were unable to program because of limitations in the platform we used to conduct the experiment.

We created a total of four different Facebook advertisements through the HTML and PHP programming options in the platform we used to conduct the experiment (SoSci Survey), meaning we changed input fields for the stimuli, which led to four different versions (i.e., pro- or anti-climate change regulations for both conditions; see procedure). For each of the two experimental conditions, the picture containing the politician was the same, the profile picture and the name of the page that posted the post were the same, as were the

number of likes and comments (see Figure 2). The statement above the post differed for the pro and anti-climate change regulations groups, leaving both groups with a statement supposedly in line with their own beliefs. The only other differences were the two disclosures that were manipulated. Our manipulated disclosures looked like the disclosures Meta/Facebook uses nowadays, although we did change the font size and color to be able to use the dynamic functions we applied (i.e., hovering and using parameters that participants reported; see Figure 3).

Figure 2

Stimuli



Figure 3

Dynamic boxes including example parameters

This message is targeted on people between the ages of 31 and 40
This message is targeted on male users
This message is targeted on you through your online behavior and interactions on Facebook

Sample

Our sample, consisting of 577 German Facebook users, was recruited through panel provider Respondi AG. From our total sample, 30 people completed the experiment in a time of less than one and a half minutes, and therefore they were excluded from the dataset, leaving a total sample size of 547 participants for the analyses. Within our sample, the age ranged from 18 to 74 years (M = 46.8, SD = 15.6). Of these participants, 199 identified as female, and 348 as male. Regarding education, most of the participants have an intermediate high school diploma (n = 163) and 123 have a university entrance qualification. 72 participants have a master's degree and 67 have a bachelor's degree. Furthermore, 57 participants had a qualifying middle school diploma and 44 had an advanced technical college entrance diploma. Finally, four participants have a doctorate and 17 participants responded that they have a different educational background.

Randomization checks showed no differences between our experimental groups concerning age ($F(1, 545) = 0.15$, $p = .736$), gender ($\chi^2(1, N = 547) = 0.14$, $p = .704$) or level of education ($\chi^2(7, N = 547) = 11.74$, $p = .109$).

Power

Due to our available resources, we aimed to collect responses from 500 participants (Lakens, 2022). However, we were able to collect 547 responses. Statistical power was calculated using R (version 4.1.2) (R Core Team, 2021) and the simsem package (Version 0.5-13) (Pornprasertmanit et al., 2021). The power analysis aimed to find the smallest effect size that could be found with a recommended minimum power of 90%, given our current sample size as well as the statistical method. The smallest effect size of interest (SESOI; Lakens et al., 2018) is complementary to and comparable to the alpha-level, i.e., a threshold for rejecting results. To determine our SESOI, we calculated different simulation analyses with the structure of the proposed structural equation model (SEM). While including different effect sizes in our simulation and keeping the α level (α = .05), power (90%), and sample size

(N = 547) constant, we ran 1,000 replications for our model. Consequently, we set our SESOI at β = |.14| and did not interpret results with effect sizes below that threshold.

Pre-tests

To at least simulate microtargeting, we intended to expose participants to advertisements in line with their beliefs. To accomplish this, we asked participants if they agreed or disagreed with a set of 10 statements regarding politically loaded topics. Four of these statements were chosen from a set of 10 pre-tested statements (N = 100), investigating whether they clearly described the views of someone who was *pro-* or *anti-climate* change regulations (i.e., someone who is *pro-climate change,* is willing to counter climate change by supporting the regulations; the opposite is true for *anti*). We showed participants a total of 10 statements on various political topics to counter test effects. The answers to the four statements regarding climate change regulations were registered and later used to show a statement that was either pro- or anti-climate change regulations on the Facebook post itself.

Moreover, in another pre-test, we tested the design and layout of our dynamic disclosure. We pre-tested four types of font styles and colors, representing different levels of prominence (N = 36), to find the one perceived as most prominent, which was finally used in our experiment.

Procedure

After providing informed consent, the participants were asked about their demographic information because we needed that information to fill the parameters for the targeting disclosure group. Thereafter, participants were asked if they agreed or disagreed with 10 statements on politically loaded topics. After the statements, participants were shortly briefed again and told that the next page would contain a Facebook post and that the 'next-button' would appear after 15 seconds, to ensure that they had a close look at the post. Thereafter, we exposed the participants to our stimuli and asked them questions regarding our

variables and our manipulation check. Finally, participants were debriefed and thanked. The average completion time was 5 minutes.

Measures

All measured constructs were tested for factor validity using confirmatory factor analyses (CFAs). The results of the CFAs and the measures of reliability and internal consistency are shown in Table 1. We measured *persuasion knowledge* through five items validated in earlier research on PMT by Kruikemeier et al. (2016) (based on Ham et al. (2015)) on a Likert scale ranging from 1 (= strongly disagree) to 7 (= strongly agree) (e.g., "The post feels like an ad"). The average variance extracted for this measure was just below the threshold of .50, but we kept all items because other reliability measures had a sufficient outcome, and the scale was validated in other research regarding PMT. We measured the *cognitive resistance* strategy 'counterarguing', and the *affective resistance* strategy 'negative affect' by asking participants to what extent they agreed with two times four statements based on van Reijmersdal et al. (2016) on a Likert scale ranging from 1 (= strongly disagree) to 7 (= strongly agree) (e.g., cognitive: "While reading I contested the information of the message" & affective: "While reading the message I felt angry"). *Source credibility* was measured using the scale by Ohanian (1990), which is often used in persuasion research and contains three dimensions (attractiveness, trustworthiness, and expertise). This scale is measured on five-point semantic scales. Asking participants which term best applies to the source of the message (i.e., the political party; e.g., "reliable – unreliable"). In the introductory text, we explicitly told the participants that these questions were about the political party. Due to an overlap in translation into German, the item for dependable – undependable for the dimension of trustworthiness was not measured in this study. The scores for the three dimensions were summed to obtain the total average score for the analyses. We measured *attitude towards politician* using a nine-item, seven-point semantic scale that has recently been used in

microtargeting research by Dobber et al. (2021). We explicitly asked participants to assess the politician portrayed in the advertisement according to the given characteristics (e.g., "corrupt – upright").

Table 1

Results of the Confirmatory Factor Analyses

Measured constructs	χ^2	*df*	*p*	CFI	TLI	RMSEA	SRMR	α	ω	AVE
Persuasion Knowledge	81.44	5	<.001	.92	.84	.17	.04	.81	.82	.48
Cognitive Resistance	71.82	2	<.001	.95	.85	.25	.03	.89	.90	.69
Affective Resistance	257.20	2	<.001	.92	.76	.48	.02	.97	.97	.89
Source Credibility	1252.00	77	<.001	.86	.83	.17	.06	.97	.97	.67
Attitude Towards Politician	315.20	27	<.001	.95	.93	.14	.03	.96	.96	.74
Targeting Knowledge	172.80	5	<.001	.92	.83	.25	.06	.89	.89	.65

Note. Reliability measures are Cronbach's α, McDonald's ω, and average variance extracted. CFI = Comparative fit index; TLI = Tucker-Lewis index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual; AVE = average variance extracted.

Exploratory variable

As an exploratory behavioral variable, we were able to measure the number of times participants in the targeting disclosure group hovered over the dynamic functions in our stimuli and thus were exposed to the actual targeting parameters we developed, which can be seen as behavioral data of participants interacting with the stimulus. We measured each

specific term (i.e., age, gender, and online behavior) and the number of times the participants hovered over it to see the specific information. Because the control condition did not have any dynamic functions, we were unable to measure this for that condition. As a second exploratory variable, we asked participants the extent to which they found the posts they received targeted at them. We did this by using the *targeting knowledge* scale developed in recent microtargeting work by Binder et al. (2022), consisting of five items measured on a seven-point Likert scale ranging from 1 (= strongly disagree) to 7 (= strongly agree) (e.g., "The post is based on my personal preferences"). Measuring this has implications for our simulation of microtargeting and its feasibility for future research.

As a manipulation check, we asked participants if they recalled a sponsored disclosure, a targeting disclosure, or no disclosure on the Facebook post. We also asked participants for their demographic information.

Results

Statistical analyses were performed using R (version 4.1.2) (R Core Team, 2021) and Jamovi (version 2.0.0.0) (The jamovi project, 2022). To investigate the differences between the control condition and the targeting disclosure condition, we analyzed our data using structural equation modeling with the variable mean scores as observed variables. Bivariate correlations, means, and standard deviations for the measured variables are shown in Table 2. The code of the conducted analyses is available on the OSF (https://osf.io/dwygm/?view_only=9e486925981f4b72ab9c16e59369687c).

Regarding our manipulation check, of the participants that were in the targeting disclosure condition, 41 recalled targeting information on the disclosure (15 of those participants used the hover function), while 124 wrongly recalled sponsored information on the disclosure, and 115 recalled no disclosure. Interestingly, of the participants who were in the control condition with the sponsored disclosure, 63 recalled seeing targeting information

in the disclosure, while 187 correctly recalled seeing a sponsored disclosure, and 17 participants did not recall the disclosure.

Table 2

Means, Standard Deviations, and Bivariate Correlations of the Measured Constructs

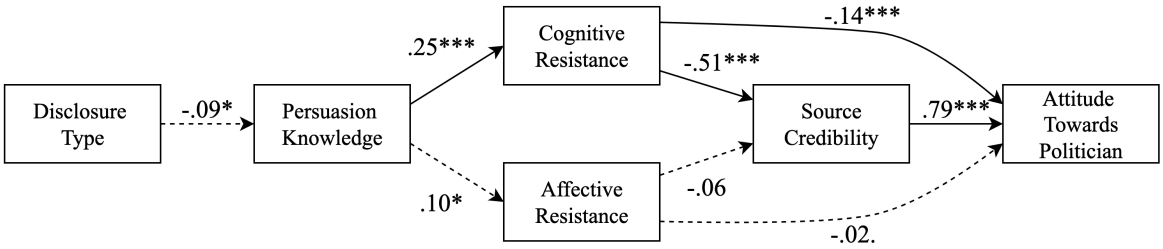
Measured construct	*M* (*SD*)	1	2	3	4	5
1 Persuasion Knowledge	4.6 (1.3)	-				
2 Cognitive Resistance	3.5 (1.7)	.25***	-			
3 Affective Resistance	2.3 (1.7)	.10*	.61***	-		
4 Source Credibility	2.6 (0.8)	-.12**	-.54***	-.37***	-	
5 Attitude Towards Politician	4.3 (1.4)	-.15***	-.57***	-.39***	.87***	-
6 Targeting Knowledge	3.3 (1.7)	-.09*	-.45**	-.22***	.67***	.65***

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. Tests to see if our data met the assumption of collinearity for source credibility and attitude towards the politician showed that multicollinearity was not a concern (Attitude towards the politician, Tolerance = .703, VIF = 1.42) (Hair et al., 2014).

The path model is shown in Figure 4. The first model did not fit well. While investigating the model fit through the modification indices, we found that cognitive resistance and affective resistance correlated, after including this correlation, our model showed an adequate fit: $\chi^2(6) = 5.99$, $p = .424$, $\chi^2/df = 0.99$, CFI = 1.00, TLI = 1.00, RMSEA = .00, 90% CI [.00, .06], SRMR = .01.

Figure 4

Path Model



Note. Numbers represent standardized regression coefficients. Dashed lines indicate a path is either statistically meaningless (not significant) or theoretically meaningless (below β = |.14|). * $p < .05$, ** $p < .01$, *** $p < .001$.

The first hypothesis (H_1) predicted that exposure to a targeting disclosure would lead to higher levels of persuasion knowledge compared to our control condition. Contrary to our hypothesis, our results revealed a small negative effect (β = -.09, $p = .022$), which is also below our SESOI (β = |.14|). Thus, we cannot interpret this effect, leading us to reject H_1. For the second hypothesis (H_2), which predicted a positive relationship between persuasion knowledge and cognitive resistance, we found a significant positive relationship (β = .25, $p < .001$), supporting this hypothesis. Furthermore, we hypothesized that persuasion knowledge would also be positively related to affective resistance (H_3). Our results showed a significant positive relationship (β = .10, $p = .031$), but the effect size is below our SESOI; therefore, we reject this hypothesis.

Our fourth hypothesis (H_4) predicted a negative relationship between cognitive resistance and source credibility. Our results showed a significant negative relationship (β = -.51, $p < .001$), supporting this hypothesis. With the fifth hypothesis (H_5), we proposed a negative relationship between affective resistance and source credibility. The results show that there is a non-significant negative relationship (β = -.06, $p = .223$). Therefore, we reject this hypothesis. For the sixth hypothesis (H_6), a negative relationship was hypothesized

between cognitive resistance and attitude towards the politician. The results show a significant negative relationship (β = -.14, p = .002), supporting this hypothesis. Furthermore, for our seventh hypothesis (H_7), a negative relationship between affective resistance and attitude towards the politician was predicted. Our results showed a non-significant negative relationship (β = -.02, p = .571), rejecting the hypothesis. Finally, our eighth hypothesis (H_8) predicted a positive relationship between source credibility and attitude towards the politician. Our results showed a significant positive relationship (β = 79, p < .001), supporting this hypothesis. In addition, we found a significant covariance between cognitive and affective resistance (β = .61, p < .001). Even though the relationships of both types of resistance on source credibility and attitude towards the politician differ, this shows that the two variables are also positively related.

As exploratory analyses (RQ1), we first investigated the differences between our measured variables based on exposure to the two different types of disclosures. We conducted ANOVAs for all measured variables with experimental groups (targeting disclosure vs. control) as factors and found significant differences for *persuasion knowledge* (F(1,545) = 5.23, p = .023), ($M_{control}$ = 4.8, SD = 1.3) ($M_{targeting}$ = 4.5, SD = 1.2) and *targeting knowledge* (F(1,545) = 9.72, p = .002), ($M_{control}$ = 3.1, SD = 1.7) ($M_{targeting}$ = 3.6, SD = 1.6). This means that for the control condition, the average level of persuasion knowledge was higher than that in the targeting condition, while for targeting knowledge, the level in the targeting condition was higher than that in the control condition. Second, we investigated whether the option to hover over terms (i.e., age, gender, online behavior) in the targeting disclosure condition was used by participants, and how this could correlate with persuasion knowledge or targeting knowledge (note: this hovering was only possible for participants in the targeting disclosure condition). Regarding the hovering, on average this possibility was used less than one time per participant (N = 280, M = 0.9, SD = 1.8). Nevertheless, only 92

participants in the experimental condition actually used the hovering option, and on average they did so between two and three times ($N = 92$, $M = 2.7$, $SD = 2.2$). We investigated differences in our measured constructs between participants that did or did not use the hovering option, we only found significant differences in *source credibility* ($F(1,278) = 8.57$, $p = .004$), ($M_{nohover} = 2.7$, $SD = 0.7$) ($M_{hover} = 2.4$, $SD = 0.7$) and *attitude towards the politician* ($F(1,278) = 5.04$, $p = .026$), ($M_{nohover} = 4.4$, $SD = 1.3$) ($M_{hover} = 4.0$, $SD = 1.4$). This means that for participants in the experimental condition who used the hovering, compared to participants in the experimental condition who did not, both source credibility and attitude towards the advertised politician were lower on average. Moreover, we investigated differences in our measured constructs between participants who used the hovering option and the control group but did not find any significant differences. Results of these analyses can be found online: (https://osf.io/dwygm/?view_only=9e486925981f4b72ab9c16e59369687c).

Discussion

The current study aimed to investigate whether targeting disclosures for political microtargeted advertisements can be beneficial for users in terms of raising their awareness that they are being targeted and how this relates to the cognitive and affective processing of advertisements as well as attitudes. To do so, we tested an integrative path model that portrays how social media users build their attitudes towards microtargeted content.

We assumed that exposure to a dynamic targeting disclosure (compared to a sponsorship disclosure), showing data used to target users on an advertisement, activates higher levels of persuasion knowledge, that is, the recognition of the message as an advertisement, in contrast to a sponsored label (Binder et al., 2022; Kruikemeier et al., 2016). Although persuasion knowledge is described as a coping mechanism regarding persuasive attempts, we found a small negative effect within our sample, which is contrary to our

expectations (Friestad and Wright, 1994; van Reijmersdal et al., 2016). However, the found relationship was below our set threshold for the SESOI; thus, the current study does not have enough power to interpret this result as meaningful. We want to emphasize that this does not have to be the case for future studies, and we encourage other researchers to investigate these potential effects. Nevertheless, we would elaborate on the negative effect on a speculative basis since it is opposite to what we predicted. Looking at the stimuli used in this study, the disclosure in the control condition clearly states that the message is sponsored and paid for by the political party. In the targeting disclosure condition, the disclosure only states that the message is targeted at the participant based on the parameters. The label does not disclose the sponsored or persuasive nature of the Facebook post. Looking at our first research question, we see that regarding persuasion knowledge, the mean in the control condition was higher than that in the targeting condition.

Consistent with previous studies, not all participants recalled the correct disclosure (Binder et al., 2022; Evans et al., 2017; Jansen and Krämer, 2023; Kruikemeier et al., 2016; van Reijmersdal et al., 2021). Only 17% of participants in the condition who saw a post with a targeting disclosure recalled the correct disclosure, while in the control condition, 70% of participants recalled the correct disclosure. Interestingly, users recall a sponsored disclosure more easily. The sponsored disclosure used in the control condition was designed to look like the one Facebook uses. Initially, we used this label to improve ecological validity while acknowledging that it has been found to not be a sufficient transparency measure; therefore, we used it as our control group (Binder et al., 2022; Binford et al., 2021). However, our findings show that the disclosure users are familiar with is recalled more easily, which means that a repeated exposure effect, increasing recognition, and potential familiarity might occur (Montoya et al., 2017). Nevertheless, we found differences between our two conditions and targeting knowledge, and argue that disclosures could improve transparency, but users might

need to become aware of targeting processes as a whole before they can recognize these processes through a disclosure. Moreover, we included hovering options in the targeting disclosure, which only 33% of participants used. However, participants who used this option used it three times on average, which shows the possibilities of more elaborate and interactive disclosures for SNS, which could take time for users to get accustomed to (Amazeen and Wojdynski, 2020; Binder et al., 2022; Boerman et al., 2015). The interactive part of the disclosure was implemented for use on a computer, but a different format is needed to be feasible for users on mobile devices (e.g., an 'I' button, providing additional targeting information if pressed on a mobile phone, or a small box with additional information if the screen is larger, such as a tablet).

Besides testing the potential effects of the targeting disclosure on persuasion knowledge, we expected that persuasion knowledge activation would relate to users resisting the message more, both cognitively and affectively (Beckert et al., 2021; van Reijmersdal et al., 2016). We observed this relationship only for cognitive resistance. Although users' persuasion knowledge did not appear to be activated by exposure to our disclosures, the relationship between persuasion knowledge and cognitive resistance implies that when users are aware that they are being persuaded, they are more likely to resist the message by counterarguing with it (Fransen, Smit, et al., 2015; Fransen, Verlegh, et al., 2015; Youn and Kim, 2019). Hence, individuals who are aware that a political message is an advertisement could have more doubts about the content of the message and rather reject it. For instance, they might think that the message is not honest and is only aimed at attracting voters. Moreover, political advertising can be considered as a high-involvement process. This is in contrast to the work by van Reijmersdal and colleagues (2016), where the product used in the stimulus material was a rather low-involvement product (i.e., a casserole mix). Political

advertising could be seen as an issue in which users experience more involvement, which might explain why they actively counterargue with the message.

The proposed relationship between persuasion knowledge and affective resistance was significant, but lower than our threshold for meaningful interpretation. This implies that users counterargue with the message, but in our sample, they might not have experienced feelings like anger to the same extent. At the same time, we argue that this finding can also be a result of our operationalization. We operationalized cognitive resistance as counterarguing, while operationalizing affective resistance as negative affect (van Reijmersdal et al., 2016). Users may not feel angry because they are targeted. While counterarguing as a psychological reaction to a message is something that users do daily, actually getting angry or annoyed might be a stronger reaction to an advertisement on a platform you receive rather passively.

Regarding users' attitudes towards the political party, we proposed that both dimensions of resistance are negatively related to users' perceived credibility of the source (Housholder and LaMarre, 2014; van Reijmersdal et al., 2016). Our analyses showed that cognitive resistance is related to a decrease in the perceived credibility of the source, implying that users who counterargue with the message perceive the source as less credible (Carr and Hayes, 2014; Deng et al., 2020; Wojdynski and Evans, 2016). However, there was no evidence regarding affective resistance; it should be noted that users may still experience affective resistance and that it could be part of other psychological processes, as it was positively correlated with cognitive resistance. An explanation for this could be that when users counterargue with a message, they actively process the information, which might lead them to engage in more critical processing of the message, a process that is more likely to be activated through disclosures (Boerman et al., 2014). In turn, this systematic and biased way of processing a message could project on users' perceptions of the party or source. For instance, users could believe that the source is less sincere because they sent a persuasive

message, potentially decreasing their credibility perceptions. In contrast, it is possible that, since there is no activation of affective resistance through persuasion knowledge, users do not have affective reactions towards the message. If this is the case, this will also not project their perceptions of the source and, thus, does not affect their perceived credibility.

In addition, we proposed that both types of resistance would have a negative relationship with attitude towards the politician, which we only found to be true for cognitive resistance. The same reasoning used to explain the relationship between both types of resistance and source credibility might explain these results. However, looking at the strengths of the relationships, we see that the direct relationship between cognitive resistance and source credibility is stronger than the direct relationship with attitude towards the politician, while the relations of affective resistance on both credibility and attitude towards the politician are low and non-significant. A reason for the difference in the strength of the relationship between cognitive resistance and credibility, as well as cognitive resistance and attitude towards the politician, could be that the politician is only advertised in the image of the post, even though the politician is fake. The source of the message, or political party, is named at the top of the post as the sender of the message and might be more visually present to some users because the politician is non-existent and therefore does not lead to the activation of prior attitudes. Moreover, the user may perceive the source of the message as the one presenting the persuasive message, while the politician is 'just' advertised in it because they are a part of that party. This might lead to negative perceptions being activated more in reaction to the party, but less towards the advertised politician.

Additionally, as expected, we found a relationship between users' perceived credibility of the source and their attitude towards the advertised politician (Funk, 1999; Madsen, 2019, p. 107; Markus, 1982). This relation could be explained as an extension of the rationale used above, while the negative perceptions towards the source might not project on

the advertised politician as much, possibly because the opposite is true as well. In that case, it is possible that when users have more favorable credibility perceptions towards a party, the politician in the advertisement benefits from that. However, this interpretation might be possible but is on a speculative basis, since we used a fake politician and a fake party, and this might have influenced our results. While the use of a fake party and politician allows research to isolate the effects of manipulations in experiments and prevent confounding of existing attitudes, we recognize that this also makes it harder to translate results to a real-world setting because of a decrease in ecological validity. For instance, the relationship between resistance and source credibility might be less strong than it could have been with an existing political party and political actor, as there would be existing attitudes that form over time (Madsen, 2019; Markus, 1982). On the contrary, it is also possible that since the political party was unknown to participants before this experiment, this led to an overestimation of our results, as participants might have had more systematic processing of the advertisement through a novelty effect, which could have led to participants thinking too much about the political party and disregarding the disclosures.

Finally, we found that cognitive and affective resistance were positively correlated, although affective resistance was not related to persuasion knowledge or source credibility. We recognize that participants saw our manipulation in an experimental setting (i.e., a timer before they could click through to the next page, forcing longer exposure), which might have led them to use more cognitive resources to process the message, including the disclosure, than they would have used in a less superficial setting. In a real-world setting, SNS users are exposed to many messages in a short amount of time, which could lead to less central processing of the messages, including the disclosures on these messages, making it harder to process transparency measures if platforms provide them (Petty and Cacioppo, 1986). Concerning disclosures as possible measures, platforms, as well as researchers, and

eventually users would benefit from a better understanding of the most effective designs concerning transparency and information (Amazeen and Wojdynski, 2020). This study shows mixed results concerning these disclosures. On the one hand, disclosures are not always recalled correctly; on the other hand, disclosures are recalled, which highlights the potential of disclosures as a transparency measure, in line with other research (Binford et al., 2021; Kruikemeier et al., 2016; van Reijmersdal et al., 2016). Furthermore, we notice that while disclosures are not always perceived and recalled in experimental studies, this might match a real-world setting. People focus more on content while scrolling Facebook timelines and perhaps less on disclosures on this content. The question arises if users consciously or unconsciously process content on their timelines, as described in the Elaboration Likelihood Model (Petty and Cacioppo, 1984). From a usage perspective, it would make sense to focus more on the actual content; for instance, users look at a video or image in their timeline before anything else (Vergara et al., 2021), which makes it even more important to ascertain that disclosures are designed fittingly: informative and salient, but not too invasive.

Altogether, our findings imply that resisting messages, as a possible result of high persuasion knowledge, is a cognitively rather than affectively shaped process. Moreover, only the cognitive processes were negatively related to both source credibility and attitude towards the advertised politician. This means that resisting persuasive attempts and attitude formation appear to be rather conscious and controlled processes that are not associated with high negative affect. Our findings underline the importance of awareness concerning microtargeting, as this seems to be a precondition for challenging the content, source, and politician of a political advertisement. Microtargeted users might resist messages more if they are aware of the targeting that occurs. In addition, this resistance possibly not only decreases perceived credibility, but also users' attitude towards the advertised politician. For political parties and advertisers working with them, this implies that using PMT should be a well-

considered choice, as it might be counterproductive for affecting voters (Hetherington, 1999; Housholder & LaMarre, 2014).

Limitations and future work

In order to not influence the participants of this study beyond this experiment, we made use of a made-up political actor and party. As described, political behavior and actual voting, while being influenced by perceived credibility and attitudes towards politicians, consist of many factors and not just exposure to online political advertisements. While avoiding ethical risks, we recognize a potential novelty effect, as participants had never seen this party or politician before. Therefore, their focus could have been more on the politician and the party, recalling the disclosure worse than they possibly would in a real-world setting, even with our 15-second timer. Future research implementing objective measures such as eye tracking would describe users' exposure to the actual disclosure better, as we had no control over that besides the timer. Furthermore, the domain of disclosures from a policy and transparency perspective would benefit from investigating whether users do not recall disclosures due to a lack of interest or an actual lack of exposure or recognition. Regarding the measures for source credibility and attitude towards the politician, there is a possibility that, even though we explicitly stated that source credibility concerned the party and attitude towards the politician concerned the actual politician, there might be overlap in these measurements. An experiment that uses existing political advertisements might be able to elaborate on the potential differences between attitudes towards a party and attitudes towards a politician.

In this experiment, we tried to simulate microtargeting by showing participants statements that were close to their own beliefs. However, we recognize that our measurements of counterarguing as a measure of cognitive resistance possibly come from users disagreeing with the statement despite our efforts to match the statement to their

previous answers. Future research can benefit from other ways of tailoring stimuli to users' beliefs while investigating PMT, for instance, by asking users more about those beliefs and using a larger battery of statements for stimuli.

In this study, we choose to compare our targeting disclosure with the disclosure that Facebook currently uses as a control group. While there is no possibility of advertising on the platform without the advertisement containing this disclosure, we still recognize that future work could scrutinize potential differences between an organic post (i.e., without a disclosure), an advertisement with the current disclosure, and a targeting disclosure to isolate potential differences. Furthermore, we recognize that the process of political communication is not isolated to the measures we used in this study, and we encourage other researchers to investigate user behavior regarding transparency and advertising on SNS through other measurements like for instance, such as ad avoidance, blocking behavior, and engagement measures such as commenting and sharing.

Moreover, to increase ecological validity, research focusing on PMT would benefit from field experiments using existing political advertising. This could be possible by having users install a browser plugin that recognizes political advertisements on their Facebook timeline to better indicate exposure. This would also have the possibility to conduct a study with a longitudinal nature, which could control for party preferences at the beginning, so possible fluctuations in political attitudes could be investigated, and the advertisements might be targeted at users to fit their beliefs and attitudes from the start. In addition, a field experiment might be able to test users' attitudes and the potential fluctuations in those attitudes due to exposure to targeted advertisements. Simultaneously, a field experiment would be able to introduce other parameters in the dynamic aspect of the disclosure, ensuring that users are aware of which different data points are used to target them, compared to our targeting parameters (age, gender, and online behavior).

Furthermore, in line with central and peripheral processing as described above, we forced participants to view the message including the disclosure for 15 seconds, while this was done to make sure they did not click through to the next page immediately, we recognize that this potentially leads to an overestimation of potential effects of exposure to a disclosure as a real-world setting might find less strong relations, due to a less forced and superficial setting. Finally, apart from the effect of the manipulation on persuasion knowledge, this study has a cross-sectional design; therefore, we are not able to investigate causal relations, meaning that the other paths in our model are bi-directional.

Conclusion

Online PMT is a technique that for most users occurs in the background. While there seems to be no consensus regarding the effects of the technique, research on transparency concerning it is also scarce. As calls for transparency regarding the use of private information for microtargeted ads have become louder, the present study investigated the effectiveness of a disclosure label to provide users with more information. We tested an integrative model that was meant to depict one possible way in which receivers of microtargeted ads might react to the content. Although the disclosure label did not have a meaningful effect on users' persuasion knowledge, persuasion knowledge was positively related to cognitive (but not affective) resistance to the ad. This shows that people begin to question the content of messages when they are aware that a political message is an advertisement. Moreover, people who challenged the content of the message attributed lower credibility to the source of the message and had a more negative attitude towards the politician. Together, these findings show that the reaction to microtargeted political content is a cognitive rather than an affective process. Although the exact role of targeting disclosures still seems ambiguous, it seems an important first step in making people aware that they are being targeted based on their personal information. Therefore, the present study is an important contribution in showing

that social media users react negatively to political microtargeting when they are aware of the use of their personal data.

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